

Supplementary Material

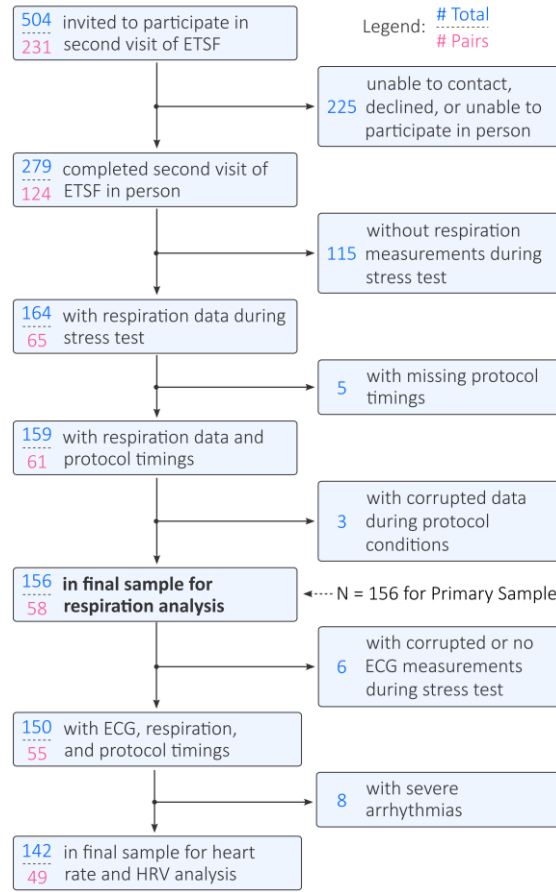


Fig. S1. Participant flow diagram. From the 504 veterans invited to participate in the second visit of the Emory Twin Study Follow-Up (ETSF), 164 completed the study in person and had respiration data collected during the trauma recall protocol. Due to missing or corrupted data, 8 of the 164 veterans were excluded, 156 veterans (58 twin pairs) of data for use in the final sample for respiration analysis. Additional veterans were removed for the supporting heart rate and heart rate variability (HRV) analyses due to corrupted electrocardiogram (ECG) measurements or severe arrhythmias.

TABLE S1
PARTICIPANT CHARACTERISTICS

Participant Characteristics	Respiration Features Sample (N = 156)	Paired Twins for Respiration (N = 116)	Cardiac Features Sample (N = 142)	Paired Twins for Cardiac (N = 98)
Monozygotic, No. (%)	93 (60)	70 (60)	85 (60)	62 (63)
Afternoon Start, No. (%)	75 (48)	58 (50)	69 (49)	49 (50)
Sociodemographic factors				
Age, y	68 (2)	68 (3)	69 (2)	68 (3)
White, No. (%)	149 (96)	108 (93)	133 (94)	90 (92)
Health factors				
BMI, Mean (SD)	29.4 (4.33)	29.4 (4.41)	29.4 (4.17)	29.3 (4.46)
History of Smoking, No. (%)	103 (66)	80 (69)	96 (68)	71 (72)
History of CVD, No. (%)	22 (14)	17 (15)	15 (11)	11 (11)
Current PTSD, No. (%)	13 (8)	9 (8)	11 (8)	8 (8)
PTSDSS Score, Mean (SD)	23.1 (9.8)	23.2 (10.4)	23.3 (9.8)	23.6 (10.4)
Medication use				
β-blockers, No. (%)	35 (22)	25 (22)	33 (23)	24 (24)

“Afternoon Start” refers to co-twins within a twin pair that participated second. Recall that both co-twins were required to participate on the same day, so one veteran started the protocol in the morning, while the other started in the afternoon. Abbreviations – BDI: Beck Depression Inventory BMI: body mass index; CVD: cardiovascular disease; PTSD: posttraumatic stress disorder; PTSDSS: PTSD Symptom Scale

TABLE S2

MODELING RESULTS FOR NEUTRAL VS. TRAUMA RECALL COMPARISONS OF PHYSIOLOGICAL MEASURES WITH NESTED RANDOM EFFECTS

Marker	β (Standard Error)	p
IQR(T_i)	0.21 (0.05)	< .001
IQR(T_e)	0.24 (0.05)	< .001
IQR(RR)	0.36 (0.06)	< .001
T_i	-0.03 (0.05)	.47
T_e	-0.11 (0.05)	.049
RR	0.18 (0.05)	< .001
Heart Rate	0.29 (0.03)	< .001
SDNN	0.15 (0.05)	.01
RMSSD	-0.05 (0.02)	.01
LF/HF	0.03 (0.07)	.62

This supplementary analysis was conducted to corroborate the neutral vs. trauma comparison detailed in the main text. This analysis leverages nested random effects to model the marker means during neutral and trauma as is, rather than consider the stress-induced differences, Δ , as the dependent variable. We fit the mixed-effects model given by $Y_{ijk} = \beta X_k + C + \alpha_i + \Omega_{ij} + \epsilon_{ijk}$. Y represents the marker of interest, β represents the fixed effect of interest (standardized as in the main text), and X represents a binary variable where $X_0 = 0$ for the neutral condition and $X_1 = 1$ for trauma recall (i.e., $k = 0$ for neutral and $k = 1$ for trauma recall). As in the main text, i indexes twin pair, and j indexes co-twin. C represents a fixed intercept, α_i represents a random intercept per twin pair, Ω_{ij} represents a random intercept per co-twin within a twin pair to account for the neutral and trauma data for the same participant in this analysis, and ϵ_{ijk} represents the model residual. Statistical procedures for this analysis matched those detailed in the main text. Bold entries indicate statistical significance after Bonferroni-Holm correction. Light blue shading corresponds to RPV markers, light purple shading corresponds to respiratory timings and rate, and light pink shading corresponds to heart rate and HRV markers.

SUPPLEMENTARY TEXT S1

Although the specific mechanisms behind the positive correlation between PTSD symptoms and Δ RPV remain unknown, the autonomic nervous system's effects on respiratory resistance may play a role. Decreased parasympathetic activity and increased sympathetic activity both decrease respiratory resistance [1]. Interestingly, a recent computational modeling study demonstrated that decreased resistance increased the variability of respiratory timings [2].

It is unclear why T_i 's variability would be most consistently related to PTSD. One possibility involves motor neuron output during inspiration and expiration. Inspiration is active: muscle contraction is always required. Expiration is passive during quiet breathing, requiring little to no motor output [3]. In this study where participants remained seated and did not speak, quiet breathing was likely the prevalent mode of respiration. Thus, inspiration was likely the respiratory phase of active pacemaker output [3], so any changes in internal rhythm would have affected T_i 's variability.

TABLE S3

STATISTICAL SYNTAX IN R (LME4 PACKAGE) CORRESPONDING TO MIXED-EFFECTS MODELS' EQUATIONS

Mathematical Notation	Equation(s) in Main Text	lmer() Model Syntax
$\Delta Y_{ij} = \beta + \alpha_i + \epsilon_{ij}$	3	$\Delta Y \sim (1 i)$
$\Delta Y_{ij} = \beta \cdot X_{ij} + C + \alpha_i + \epsilon_{ij}$	4, 5	$\Delta Y \sim X + (1 i)$
$\Delta Y_{ij} = \beta \cdot X_{ij} + C + \alpha_i + \epsilon_{ij} + \sum_{k=1}^K \gamma_k \cdot Z_k$	6	$\Delta Y \sim X + Z_1 + Z_2 + \dots + Z_K + (1 i)$
$\Delta Y_{ij} = \beta \cdot (X_{ij} - \bar{X}_i) + \gamma \cdot \bar{X}_i + C + \alpha_i + \epsilon_{ij}$	7	$\Delta Y \sim X_w + X_b + (1 i)$
$Y_{ijk} = \beta \cdot X_k + C + \alpha_i + \Omega_{ij} + \epsilon_{ijk}$	N/A (see Table S2)	$Y \sim X + (1 i) + (1 i:n)$

Y and X represent the dependent (e.g., RPV) and independent (e.g., PTSDSS) variables of interest, respectively. $Z_1, Z_2, \dots, Z_K$ represent the K covariates that are adjusted for. X_w stores the difference between each co-twin's X and the twin pair's average X value, X_b . The Δ symbol is used to denote the difference between trauma and neutral, as is detailed in the main text. The i variable represents a factor that assigns the same value to twins of the same twin pair (i.e., co-twins), while j represents a binary variable that differs between co-twins. The variable n represents a factor storing participant number, ranging from $n = 1, 2, \dots, N$, where N is the total number of participants included in the analysis. Note that N is equivalent to the total number of ordered pairs (i, j) in the dataset.

SUPPLEMENTARY REFERENCES

- [1] V. H. J. Van Der Velden and A. R. Hulsman, "Autonomic Innervation of Human Airways: Structure, Function, and Pathophysiology in Asthma," *Neuroimmunomodulation*, vol. 6, no. 3, pp. 145–159, 1999.
- [2] J. Jaworski and J. H. T. Bates, "Sources of breathing pattern variability in the respiratory feedback control loop," *J. Theor. Biol.*, vol. 469, pp. 148–162, May 2019.
- [3] S. R. Braun, "Respiratory Rate and Pattern," *Clin. Methods Hist. Phys. Lab. Exam.*, 1990.